



ME 207: Fluid Dynamics

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Project Title:- Liquid Sloshing in a tank subjected to rectilinear acceleration

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Group 14

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1 Introduction

Liquid sloshing refers to the oscillatory motion of a liquid's free surface inside a partially filled container when subjected to acceleration. This phenomenon occurs because of the liquid's inertial response to the motion of the container. That creates a wave-like motion on the surface of the liquid. The motion of the liquid generates forces on the container walls and changes the position of the center of mass to influence the system's stability. During the liquid sloshing in a tank, the liquid's free surface makes an angle with the horizontal axis, which is called the liquid sloshing angle. The liquid sloshing angle responds to the container's acceleration, vibration or motion.

Understanding liquid sloshing is important in many engineering applications, such as transporting or storing liquids. Sloshing can affect the safety, stability, structural integrity, and efficiency of many industries, such as automotive, aerospace, marine, etc. For example, if tankers carry water, fuel or chemicals, sloshing can affect the brakes or turning, which can cause an accident.

This project aims to find the relationship between the liquid sloshing angle and the tank's acceleration and check whether a fluid's viscosity and density affect the sloshing angle. For that, we designed an experimental setup with a small tank mounted on the cart (made up of an MDF sheet and wheels). Moreover, we generate constant acceleration using the free-falling weights to analyze the liquid sloshing behavior.



Figure 2: Images of the experimental setup.

2 Importance of Liquid Sloshing

- Sloshing exerts pressure on the container wall, which must be considered for the structural strength. This pressure interaction is important in designing containers or any other structure holding liquid.
- Sloshing helps to explain certain natural phenomena, like the movement of water inside a reservoir during an earthquake or the motion of the magma in a volcano.
- Sloshing has some practical uses also, such as in washing machines; sloshing helps in cleaning by enhancing mixing and is also used in wave energy devices to measure energy from ocean waves.
- Liquid sloshing is used in aerospace industries, where liquid sloshing affects rocket fuel delivery. In marine systems, it affects the ship's movement and stability.

3 Procedure

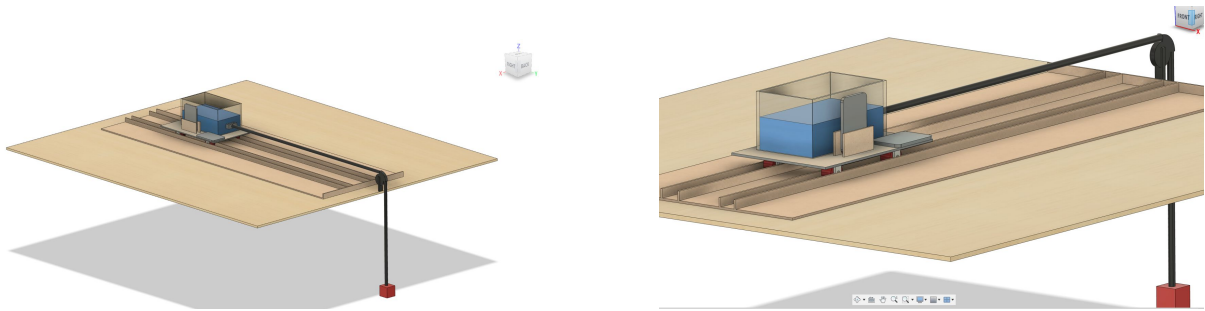


Figure 3: Images of the proposed experimental setup.

- We measured the sloshing angle with the help of the free fall of the weight. Above, we can see the experimental setup for this project.
- There are three rails (each has a boundary so the wheels trolley can't distract from the track), one trolley, one free weight and one pulley in the experimental setup. The free weight is attached to the trolley with the help of rope throughout the pulley.
- The trolley is kept on the rails. This whole setup was set to a certain height (around 2 meters).
- We kept the trolley at the end of the rail so that we got maximum time to observe the sloshing of the fluid.
- We waited for the fluid to be stable. After that, we dropped the free weight to initiate the motion.
- We assumed only the horizontal acceleration of the trolley in this motion. Because of the motion, the fluid in the trolley made an angle with the horizontal. One phone recorded the video of the sloshing and tracked the state where the fluid made a perfect and steady sloshing angle, which we measured. Another phone was kept at the top of the trolley, and the acceleration was measured with the help of an accelerometer.

- We used measurement scales, which are drawn in the trolley, to measure the sloshing angle.
- From the value of acceleration, we calculated the value of the sloshing angle corresponding to that acceleration ($\theta = \tan^{-1}(\frac{a}{g})$) and then compared it with the measured sloshing angle. We took three readings to ensure that the comparison is justifiable.
- The same procedure can be repeated for the different fluids.

4 Calculations and Results

4.1 Derivation

Below is a detailed analysis showing the relationship between the sloshing angle and the acceleration.

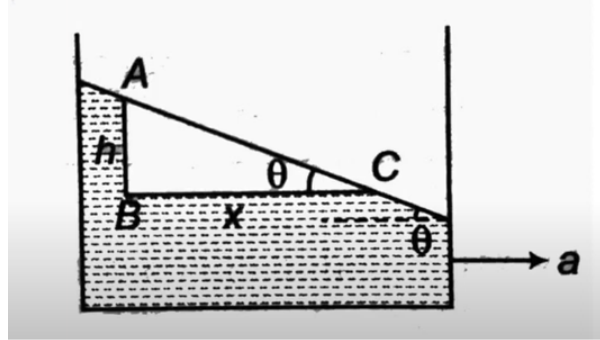


Figure 4: Schematic image for derivation

From the diagram above, pressure at point B can be written in two ways.

$$P_B = P_C + \rho ax = P_A + \rho gh$$

- P_A, P_B, P_C are the pressures at point A, B and C respectively.
- ρ is the density of fluid,
- g is the acceleration due to gravity,
- a is the rectilinear acceleration generated using free weight.
- θ is the sloshing angle.

The term ρgh arises because of the vertical pressure gradient, while the term ρax arises because of the horizontal pressure gradient.

Since A and C are exposed to atmosphere, we can reduce the above equation to

$$\rho ax = \rho gh$$

Thus,

$$\frac{h}{x} = \frac{a}{g}$$

Now, the term $\frac{h}{x}$ is nothing but the tangent of the sloshing angle so we can write it as

$$\tan(\theta) = \frac{h}{x} = \frac{a}{g}$$

Thus the sloshing angle θ is given as

$$\theta = \tan^{-1}\left(\frac{a}{g}\right)$$

4.2 Data

As mentioned in the video, we did the experiment with two different fluids: cooking oil and water.

By using the known value of their density and dimensions of the container (which are given below), we could compute the mass of the liquid inside the container.

Dimensions of the container

- $L = 19 \text{ cm}$ (Length of the container)
- $B = 15.5 \text{ cm}$ (Breadth of the container)
- $H = 22 \text{ cm}$ (Height of the container)

Densities of the fluid

- $\rho_o = 0.90 \text{ g/cm}^3$
- $\rho_w = 1 \text{ g/cm}^3$

4.3 Results

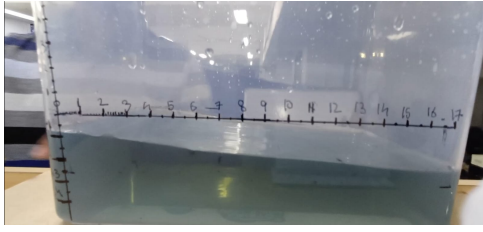
After performing the experiment, we got the video recorded on the phone; we took a snapshot of when the profile looked somewhat steady. We could measure the sloshing angle with the help of markings made on the container. Apart from that, we also got the value of acceleration measured by the accelerometer present in the phone, from which we calculated the value of the sloshing angle corresponding to that acceleration, given by the formula stated above ($\theta = \tan^{-1}\left(\frac{a}{g}\right)$).

We calculated the sloshing angle for two different volumes of water in the container to see whether the sloshing angle was affected by the density or the viscosity; we also performed it by keeping cooking oil in the container.

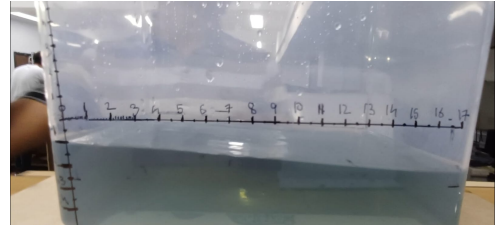
Detailed Analysis of all three cases is done below.

For water (Volume = 1089 cm^3 , Mass = 1089 g)

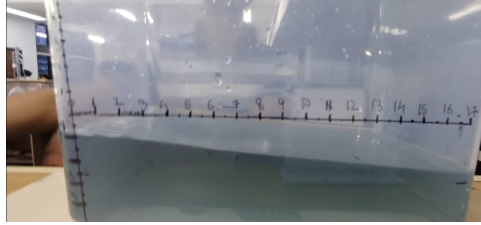
The images shown below are of the surface profiles in the case of water for a volume of 1089 cm^3 .



(a) Take 1



(b) Take 2



(c) Take 3

Figure 5: Profiles for the first case

Here are the values of the sloshing angle calculated manually using the markings and the ones calculated using the value achieved from the accelerometer for the first case.

Reading no.	θ_{acc}	θ_m
1	7.14	6.4
2	6.4	5.87
3	6.83	6.4
Average	6.79	6.22

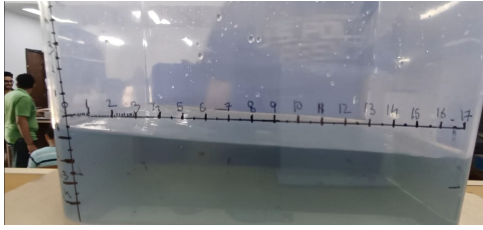
Table 1: Readings for the first case

- θ_{acc} is the value of the sloshing angle corresponding to the acceleration measured by the accelerometer (in the mobile phone).
- θ_m is the measured value of the sloshing angle, which we calculated using the markings done on the container.

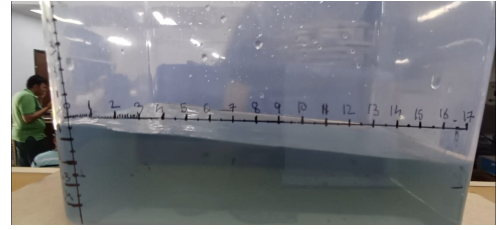
All values are in degrees.

For water (Volume = 1236 cm^3 , M = 1236 g)

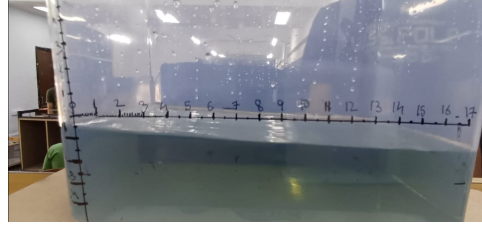
The images shown below are of the surface profiles in the case of water for a volume of 1236 cm^3 .



(a) Take 1



(b) Take 2



(c) Take 3

Figure 6: Profiles for the second case

Here are the values of the sloshing angle calculated manually using the markings and the ones calculated using the value achieved from the accelerometer for the second case.

Reading no.	θ_{acc}	θ_m
1	5.24	5.88
2	5.82	4.84
3	5.29	5.7
Average	5.45	5.47

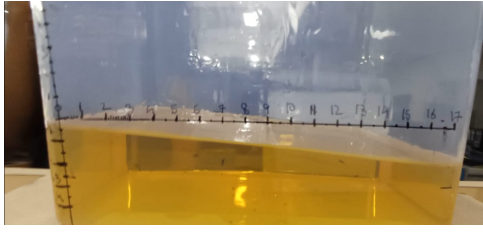
Table 2: Readings for the second case

- θ_{acc} is the value of the sloshing angle corresponding to the acceleration measured by the accelerometer (in the mobile phone).
- θ_m is the measured value of the sloshing angle, which we calculated using the markings done on the container.

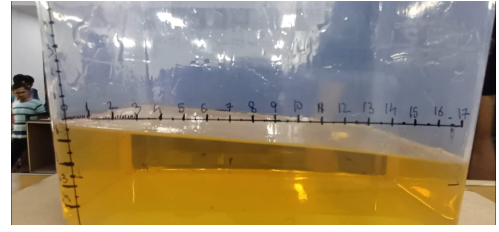
All values are in degrees.

For Oil (Volume = 1236cm^3 , Mass = 1124 g (Close to the mass of water in the first case))

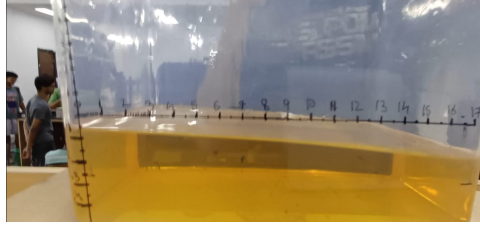
The images shown below are of the surface profiles in the case of oil for a volume of 1236cm^3 .



(a) Take 1



(b) Take 2



(c) Take 3

Figure 7: Profiles for the third case

Here are the values of the sloshing angle calculated manually using the markings and the ones calculated using the value achieved from the accelerometer for the third case. It was verified that the mass of oil in this case is nearly the same as the mass of water in the first case. So, the effects of density and viscosity can be analyzed.

Reading no.	θ_{acc}	θ_m
1	5.88	6.45
2	5.77	5.92
3	5.82	5.38
Average	5.82	5.92

Table 3: Readings for the third case

- θ_{acc} is the value of the sloshing angle corresponding to the acceleration measured by the accelerometer (in the mobile phone).
- θ_m is the measured value of the sloshing angle, which we calculated using the markings done on the container.

All values are in degrees.

4.4 Comparison of measured sloshing angle with the one predicted using the value from the accelerometer.

Here is the summarized table showing the comparison between the sloshing angle observed and the one predicted by the value of acceleration from the accelerometer. We have calculated the percentage deviation using the below expression

$$\%deviation = \left| \frac{\theta_{acc} - \theta_m}{\theta_{acc}} \right| \times 100\%$$

Case	θ_{acc}	θ_m	%deviation
1	6.79	6.22	9.16%
2	5.45	5.47	0.36%
3	5.82	5.92	1.68%

Table 4: Summarized data and deviation

Here, we can observe that the value of the sloshing angle calculated using the value of acceleration recorded by the accelerometer in the mobile phone is nearly the same as the sloshing angle observed.

5 Error Analysis

5.1 Calculation of error

For the calculation of error, we took the mean of the readings and took the difference of individual readings from the mean. This way, we calculated the average error.

Thus, the error propagated in the sloshing angle is given by

$$E = \frac{1}{n} |\theta_{mean} - \theta|$$

- θ_{mean} is the average sloshing angle.
- n is the number of readings taken
- E is the error propagated.

This has been done for all three cases

For the first case ($\theta_{mean} = 6.22$)

Reading no.	θ	$ \theta_{mean} - \theta $
1	6.4	0.18
2	5.47	0.35
3	6.4	0.18
Average	6.22	0.26

Table 5: Error for the first case

All values are in degrees.

For the second case ($\theta_{mean} = 5.47$)

Reading no.	θ	$ \theta_{mean} - \theta $
1	5.88	0.41
2	4.84	0.63
3	5.70	0.23
Average	5.47	0.42

Table 6: Error for the second case

All values are in degrees.

For the third case ($\theta_{mean} = 5.92$)

Reading no.	θ	$ \theta_{mean} - \theta $
1	6.45	0.53
2	5.92	0.00
3	5.38	0.54
Average	5.92	0.35

Table 7: Error for the third case

All values are in degrees.

Now, since we have the mean value of the sloshing angle and average error, we can calculate the percentage error for each case. The percentage error is given by

$$\%E = \frac{\Delta\theta_{mean}}{\theta_{mean}} \times 100\%$$

- $\Delta\theta_{mean}$ is the average error in the measured sloshing angle.
- θ_{mean} is the average sloshing angle

The percentage error in the sloshing angle measure for each case is given by

Case	θ_m	$\Delta\theta$	$\%Error$
1	6.22	0.26	4.18%
2	5.47	0.42	7.6%
3	5.92	0.35	5.9%

Table 8: Summarized data and percentage error

5.2 Sources of error

1. **Usage of Ultrawide Camera** - We recorded the video of the container using the ultrawide camera present on the mobile phone. This leads to some distortions, such as the vertical edge of the container not looking vertical, which leads to an error in the measured value.
2. **Friction** - We expected acceleration to be constant from the free-body analysis, but because of non-uniform friction, the value of acceleration measured by the accelerometer was fluctuating.
3. **Response Error** - To compare the measured sloshing angle with the calculated sloshing angle (using the accelerometer), the accelerometer and the video were required to start at the same time. Since this task was done manually, there may be some delay, which can cause uncertainties.
4. **Disturbances** - To ensure a steady sloshing angle, acceleration was required to be unidirectional; however, some disturbances, such as minor shaking of the table, can contribute to the error in the measured value.

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5. **Setup inaccuracies** - We did this experiment on four tables, the small error in the table's alignment also affected the motion of the container. It affects the acceleration and the liquid sloshing angle.
 6. **Orientation of Wheels** - The orientation of the wheels of the cart did not remain the same all the time; the wheels steered at some locations, which affected the motion of the cart.

6 Key observations and Conclusion

- It was observed that for nearly the same mass of oil and water, the sloshing angle does not change much. The mass of oil in the container corresponding to the third case (Volume = 1236cm^3 , Mass = 1124 g) is nearly the same as that of the mass of water in the container in the first case (Volume = 1089 cm^3 , Mass = 1089 g), and the observed sloshing angles are 5.92° and 6.22° , respectively. This implies that density and viscosity do not have a significant effect on the sloshing angle, which was expected from the theory.
- The observed surface profile turned out to be better in the case of cooking oil than in the case of water. The profile was more linear, and it was easier to measure the sloshing angle.

7 References

- [1]“Liquid Sloshing - an overview | ScienceDirect Topics,” [www.sciencedirect.com](https://www.sciencedirect.com/topics/materials-science/liquid-sloshing).
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- [2] Doubtnut, “A liquid kept in a container has a ...,” YouTube. Nov. 07, 2021. [Online]. Available: <https://www.youtube.com/watch?v=AkJpdcwZ2x4>
- [3]“What Is Sloshing? The Dangers of Tank Slosh,” Arnold & Itkin LLP. <https://rb.gy/rigd1a>